TECHNIQUE MASTERING EXERCISES

H - LIMITS

1.	does not exist since	**2.**	does not exist	**3.**	6
4.	-1	**5.**	$\frac{\pi}{2}$	**6.**	0
7.	2	**8.**	does not exist	**9.**	0
10.	17	**11.**	∞	**12.**	0
13.	$-\infty$	**14.**	$\frac{-1}{\sqrt{3-y}}$	**15.**	$-\infty$

Example number 2

$$\lim_{x\to 1} \frac{x^2-1}{x^2-2x+1} = \lim_{x\to 1} \frac{(x-1)(x+1)}{(x-1)(x-1)} = \lim_{x\to 1} \frac{(x+1)}{(x-1)}$$

$\lim_{x\to 1^+} \frac{(x+1)}{(x-1)} = +\infty$ and $\lim_{x\to 1^-} \frac{(x+1)}{(x-1)} = -\infty$ and therefore the limit does not exists.